

Kaushlendra Kumar

Curriculum Vitae

School of Mathematical Sciences
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Profile

- Research** My research lies in mathematical and theoretical physics, connecting exact gauge fields, quantum/noncommutative geometry, spectral distance, quantum information geometry, and emergent spacetime. Recent work computes black-hole quantum-geometric flow and phase diagnostics, finite spectral distances with quantum-information meaning, exact Yang–Mills/coset fields and source/transport laws, finite Clifford spectral triples, and matrix-model corrections to gravitational dynamics.
- Keywords** Quantum geometry, exact Yang–Mills fields, noncommutative geometry, spectral distance, quantum information geometry, black-hole transport, matrix/emergent gravity.
- Identifiers** ORCID: 0000-0001-5665-575X; arXiv: [kumar_k_2](#); InspireHEP: 1706741.
- Publication summary** 10 refereed journal articles, 2 conference proceedings, and 4 current public preprints.
- Research website** k-kumar.netlify.app/work: public research map, papers/preprints, talks, public engagement, and updated programme status.

Positions

- Mar 2024–Nov 2027 **DFG Walter-Benjamin Research Fellow**, *School of Mathematical Sciences*, Queen Mary University of London, Quantum Riemannian Geometry with Shahn Majid.
- Jul 2023–Feb 2024 **Postdoc**, *Institute of Physics*, Humboldt University of Berlin, Germany. Funded by the Volkswagen Stiftung grant of Nichol Furey.
- 2023–2024 (4 months) **Junior Research Fellow**, *Erwin Schrödinger Institute for Mathematics and Physics*, University of Vienna.
- May 2022–Jun 2023 **Postdoc**, *Institute for Theoretical Physics*, Leibniz University Hannover (ITP-LUH), Germany. Funded by DAAD and ITP-LUH.

Major Fellowships

- 2024–2027 **Deutsche Forschungsgemeinschaft (DFG)**, Walter-Benjamin Fellowship, Queen Mary University of London.
- 2018–2022 **Deutscher Akademischer Austauschdienst (DAAD)**, Doctoral scholarship, Leibniz University Hannover.
- 2016–2018 **Council for Scientific and Industrial Research (CSIR)**, Junior Research Fellowship (JRF), IISER Kolkata.

Education

- Oct 2018 – May 2022 **PhD Theoretical Physics**, *Institute for Theoretical Physics*, Leibniz University Hannover, Germany.
- Thesis title Solutions of Yang–Mills theory in four-dimensional de Sitter space
- Grade Magna cum Laude
- Defense date 4 May, 2022

- Jun 2018 – Sep 2018 **German language**, *Carl Duisberg Centrum Berlin*, mandatory pre-PhD language course funded by DAAD, Level reached: B1 (Intermediate).
- Jul 2015 – May 2018 **MSc Physics**, *Department of Physical Sciences*, Indian Institute of Science Education & Research, Kolkata (IISER-K), India.
Grade 9.25/10
Defense date 20 Jul, 2017
- Jul 2010 – May 2015 **BTech & MTech Biochemical Engineering**, *School of Biochemical Engineering*, Indian Institute of Technology, Banaras Hindu University (IIT-BHU), Varanasi, India.
Grade 7.88/10
Defense date 27 May, 2015

List of publications

(a) Regular articles

- 2024 **Kaushlendra Kumar** and Harold C. Steinacker. Modified Einstein equations from the 1-loop effective action of the IKKT model. *Classical and Quantum Gravity*, volume 41, page 185007, 2024. DOI:10.1088/1361-6382/ad6e4b.
- 2024 Savan Hirpara, **Kaushlendra Kumar**, Olaf Lechtenfeld, and Gabriel Picanço Costa. Exact gauge fields from anti-de Sitter space. *Journal of Mathematical Physics*, volume 65, page 072903, 2024. DOI:10.1063/5.0150027.
- 2022 **Kaushlendra Kumar**, Olaf Lechtenfeld, Gabriel Picanço Costa, and Jona Röhrig. Yang–Mills solutions on Minkowski space via non-compact coset spaces. *Physics Letters B*, volume 835, page 137564, 2022. DOI:10.1016/j.physletb.2022.137564.
- 2022 **Kaushlendra Kumar**, Olaf Lechtenfeld, and Gabriel Picanço Costa. Trajectories of charged particles in knotted electromagnetic fields. *Journal of Physics A: Mathematical and Theoretical*, volume 55, page 315401, 2022. DOI:10.1088/1751-8121/ac7c49.
- 2022 Lukas Hantzko, **Kaushlendra Kumar**, and Gabriel Picanço Costa. Conserved charges for rational electromagnetic knots. *The European Physical Journal Plus*, volume 137, page 407, 2022. DOI:10.1140/epjp/s13360-022-02563-4.
- 2021 **Kaushlendra Kumar**, Olaf Lechtenfeld, and Gabriel Picanço Costa. Instability of cosmic Yang–Mills fields. *Nuclear Physics B*, volume 973, page 115583, 2021. DOI:10.1016/j.nuclphysb.2021.115583.
- 2020 **Kaushlendra Kumar** and Olaf Lechtenfeld. On rational electromagnetic fields. *Physics Letters A*, volume 384, page 126445, 2020. DOI:10.1016/j.physleta.2020.126445.
- 2018 **Kaushlendra Kumar** and Biswajit Chakraborty. Spectral distances on the doubled Moyal plane using Dirac eigenspinors. *Physical Review D*, volume 97, page 086019, 2018. DOI:10.1103/PhysRevD.97.086019.
- 2018 Yendrembam Chaoba Devi, **Kaushlendra Kumar**, Biswajit Chakraborty, and Frederik G. Scholtz. Revisiting Connes’ finite spectral distance on noncommutative spaces: Moyal plane and fuzzy sphere. *International Journal of Geometric Methods in Modern Physics*, volume 15, page 1850204, 2018. DOI:10.1142/S0219887818502043.
- 2015 **Kaushlendra Kumar**, Shivraj Prajapat, and Biswajit Chakraborty. On the role of Schwinger’s SU(2) generators for Simple Harmonic Oscillator in 2D Moyal plane. *The European Physical Journal Plus*, volume 130, page 120, 2015. DOI:10.1140/epjp/i2015-15120-7.

(b) Conference proceedings

- 2023 **Kaushlendra Kumar**. SO(1,3) Yang–Mills solutions on Minkowski space via cosets. *SciPost Physics Proceedings*, volume 14, page 033, 2023. DOI:10.21468/SciPostPhysProc.14.033.

- 2023 **Kaushlendra Kumar**. On Yang–Mills fields from anti-de Sitter spaces. *Journal of Physics: Conference Series*, volume 2667, page 012018, 2023. DOI:10.1088/1742-6596/2667/1/012018.

(c) Preprints

- 2026 **Kaushlendra Kumar** and Shahn Majid. Geometric Aharonov–Bohm phase effect around a black hole. *arXiv preprint*, 2026. arXiv:2605.28582 [gr-qc].
- 2026 **Kaushlendra Kumar** and Shahn Majid. Geodesic flows on a black-hole background. *arXiv preprint*, 2026. arXiv:2603.03222 [gr-qc].
- 2026 **Kaushlendra Kumar**. Characteristic Lightcone Sources in $SO(1,3)$ Yang–Mills Theory. *arXiv preprint*, 2026. arXiv:2606.27241 [hep-th].
- 2026 **Kaushlendra Kumar**. Calibrated Helstrom geometry on the Bloch ball via Connes spectral distance. *arXiv preprint*, 2026. arXiv:2606.13824 [quant-ph].

Professional service

Referee Journal of Geometry and Physics.

Teaching

- Student collaboration Guided calculations with student collaborators in exact-field projects at Hannover, connecting their computations to joint papers with senior collaborators.
- Spring, 2020 **Introduction to Quantum Theory, intensive 1-week revision course**, ITP-LUH.
- Fall, 2019 **Mathematical Methods of Physics, intensive 1-week revision course**, ITP-LUH.
- Spring, 2019 **Lie groups, Algebra and their Representations, teaching assistant**, ITP-LUH.
- Fall, 2017 **Advanced Mathematical Methods of Physics, teaching assistant**, IISER-K.
- Spring, 2017 **Intermediate Electricity and Magnetism, teaching assistant**, IISER-K.

Selected talks

- 1 Jul, 2026 **XLIII Workshop on Geometric Methods in Physics, Białystok**.
Scalar-anchored Connes distance and Helstrom geometry of the qubit Bloch ball (conference talk).
- 9 Dec, 2025 **Quantum Algebras seminar, Queen Mary University of London**.
Geodesic flows on a black-hole background.
- 27 Jul, 2023 **QTS12 conference, Prague**.
Exact gauge fields from anti-de Sitter space.
- 4 May, 2023 **Erwin Schrödinger Institute, Vienna**.
Exact gauge fields from anti-de Sitter space.
- 15 Mar, 2023 **Solitons@Work**.
Exact gauge fields from anti-de Sitter space (online).
- 23 Aug, 2022 **String Theory Journal Club, DESY Hamburg**.
Yang–Mills solutions on Minkowski space via non-compact coset spaces.
- 18 Jul, 2022 **GROUP34 conference, Strasbourg**.
Yang–Mills solutions on Minkowski space via non-compact coset spaces.
- 16 Mar, 2022 **Solitons@Work**.
Yang–Mills solutions on Minkowski space via non-compact coset space (online).
- 7 Jul, 2017 **S.N. Bose National Centre for Basic Sciences, Kolkata**.
Spectral distances on doubled Moyal plane.

Schools/Conferences

- 15–17 Jul, 2026 **Black Hole Horizons and Beyond**, New Frontiers in Gravity and Holography, Durham, United Kingdom.
upcoming
- 29 Jun–4 Jul, 2026 **XLIII Workshop on Geometric Methods in Physics**, Białystok, Poland.
talk given on 1 Jul
- 07–11 Jul, 2025 **A Quantum of Noncommutativity**, Workshop/conference in Ambleside, United Kingdom.
- 24–28 Jul, 2023 **QTS12**, The 12th International Symposium on Quantum Theory and Symmetries at Prague, Czech Republic.
- 06–17 Feb, 2023 **Vort2023**, Vortex Moduli workshop at ICTS, Bangalore, India.
online participation
- 29 Aug–9 Sep, 2022 **Saalburg28**, The 28th W.E.Heraeus Saalburg Summer School for Graduate Students focused on "Modern methods in quantum field theory" at Bayrischzell, Germany.
- 18–22 Jul, 2022 **GROUP34**, The 34th International Colloquium on Group Theoretical Methods in Physics at Strasbourg, France.
- 30 Aug–10 Sep, 2021 **Saalburg27**, The 27th W.E.Heraeus Saalburg Summer School for Graduate Students focused on "Gravitational waves and Black holes" at Heigenbruecken, Germany.
- 02–07 Aug, 2021 **ICMP**, International Congress on Mathematical Physics preceded by Young Researchers Symposium at Geneva, Switzerland.
- 31 Aug–11 Sep, 2020 **Saalburg26**, The 26th W.E.Heraeus Saalburg Summer School for Graduate Students focused on "QFT, Integrable systems and String field theory" at Heigenbruecken, Germany.
- 23–27 Sep, 2019 **SHGP**, Summer School on the topic of "Special Holonomy – Geometry and Physics" at Freiburg, Germany.
- 02–13 Sep, 2019 **Saalburg25**, The 25th W.E.Heraeus Saalburg Summer School for Graduate Students focused on "Asymptotic symmetries and Hamiltonian formulation of Gravity" at Heigenbruecken, Germany.
- 15–19 Jan, 2018 **QGNCG**, Conference on "Quantum Groups & Noncommutative Geometry" at National Institute of Science Education & Research (NISER), Bhubaneswar, India.
- 06–24 Nov, 2017 **GGD2018**, Preschool + Advanced school on the topic of "Geometry, Groups and Dynamics" at International Centre for Theoretical Sciences (ICTS) Bangalore, India.

Language skills

- Indic Maithili, Hindi and Bengali.
- English Fluent.
- German Intermediate level (B1).

Research computation

- Tools Mathematica/Wolfram Language, $\text{\LaTeX}$, GnuPlot, symbolic and numerical computation.
- Methods Finite-dimensional spectral-geometry calculations, exact tensor/curvature computations, Yang–Mills mode algebra, PDE/transport postprocessing.
- Systems Linux, macOS, Windows.